S&P GLOBAL & CRISIL CAMPUS HACKATHON 2026

Foreshock

Unstructured news and social text in. Structured risk signals out, driving a tactical index rebalancer and an event-driven stress tester.

Nael Saade· ABV

AI/NLP Risk Engine · Module A: tactical rebalancer · Module B: strategic stress testing

SIGNAL TAPE · REPLAY WEEK

8.7	Central-bank surprise hike	-4.89%
9.1	Gulf shipping standoff	-0.67%
9.5	Regional bank failure	-1.62%
8.6	Category-4 hurricane	-0.53%
8.1	Payments ransomware	-0.27%
6.2	Unconfirmed rumour	*held back*

Chip = impact score (1-10). Right = stress P&L on an 85-position wholesale book. All data synthetic.

Risk signals hide in unstructured text

The business problem

Desks and risk teams cannot read thousands of headlines and posts an hour. They need a structured, trustworthy signal: how positive or negative it is, what kind of event it describes, and how big it could be.

1. **Volume and velocity**
   News wires, GDELT-style feeds and social posts never stop.

2. **Noise and trust**
   Reposts and rumours must not move money; wire-confirmed facts should.

3. **Two time horizons**
   Intraday tilts for an index; scenario stress for a wholesale book.

Approach: sense, score, act



Sense

Ingest news and social posts, drop reposts, link each item to an S&P-100 name or the broad market.



Score

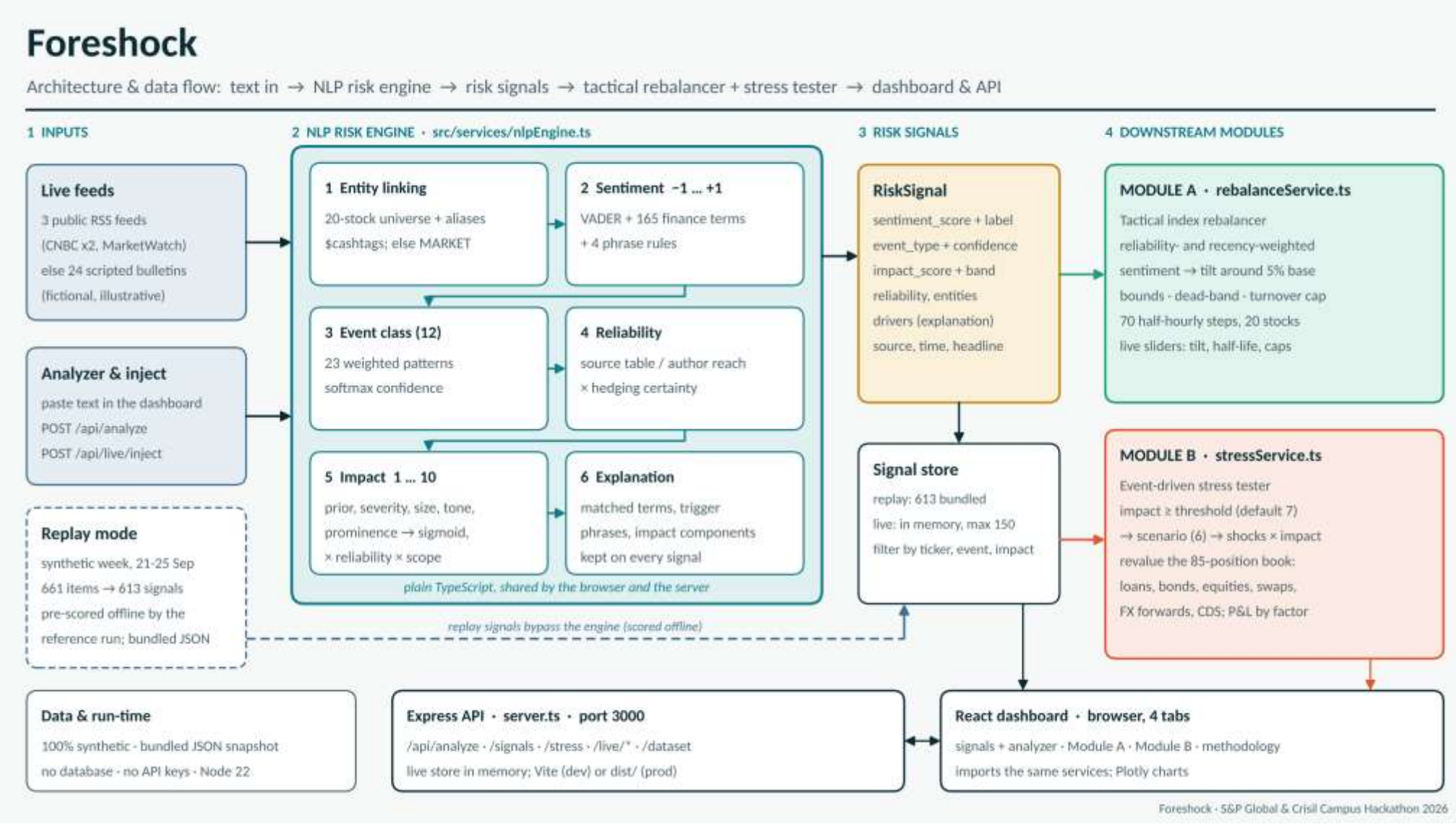
Sentiment from −1 to +1, one of 12 event classes, impact 1–10, damped by source reliability and corroboration.



Act

Module A rebalances a 20-stock index. Module B stress-tests an 85-position wholesale book.

System design: one engine, two consumers



Implementation highlights

CORE

AI/NLP Risk Engine

- VADER plus a finance lexicon, phrase and direction rules
- 12 event classes from weighted evidence, with confidence
- Impact 1-10 from event prior, severity cues, size and prominence
- Damped by source credibility, hedging and corroboration

≈1,400 items per second on one CPU core

MODULE A

Tactical rebalancer

- Per-stock sentiment: decayed, shrunk, credibility-weighted
- Tilt base weights, then bounds, dead-band and turnover cap
- Market-wide bad news de-risks high-beta names
- No look-ahead; every move traces to its headlines

83% less trading than a naive version

MODULE B

Strategic stress tester

- Fires on impact above 7, adverse tone, mapped event class
- Shocks scaled by impact: equity, rates, spreads, PD, FX
- Revalues loans, bonds, equities, swaps, FX forwards, CDS
- P&L attribution by risk factor reconciles exactly

6 scenarios · 85 positions · 3 derivative types

Key results: the engine and Module A

613

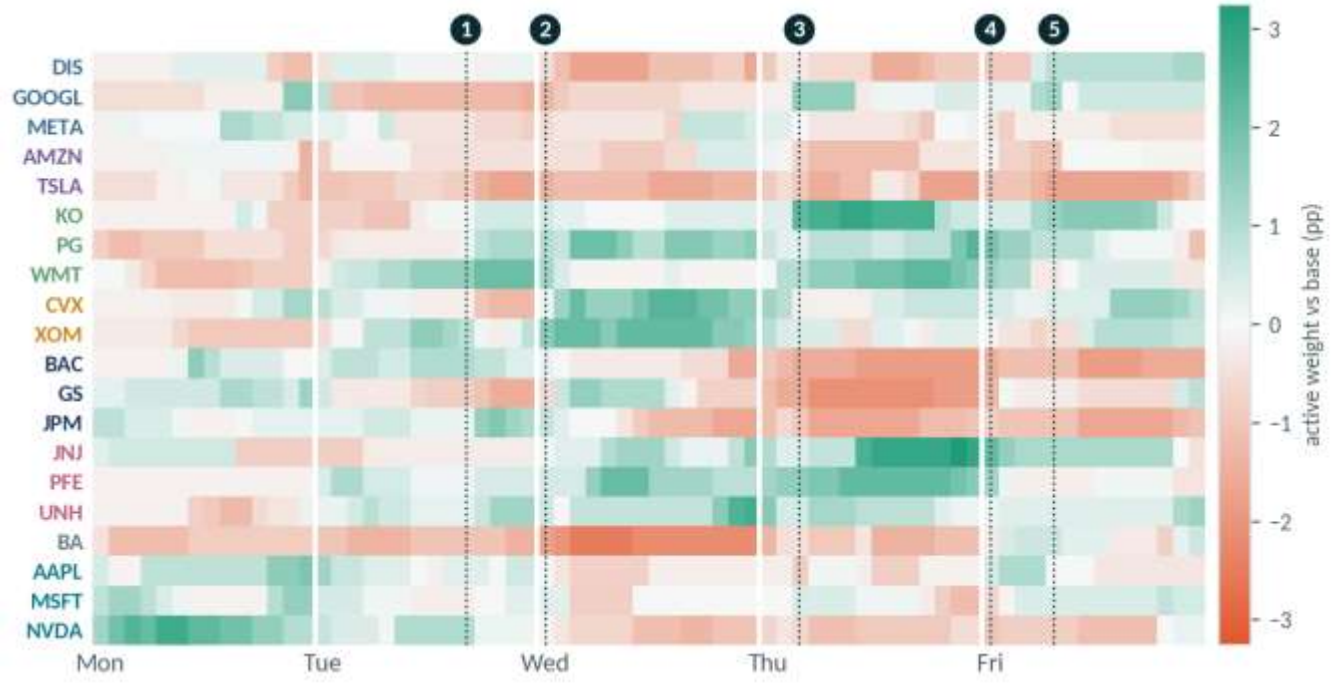
signals from 661 news and social items in under 0.5 s; 48 reposts removed

0.94

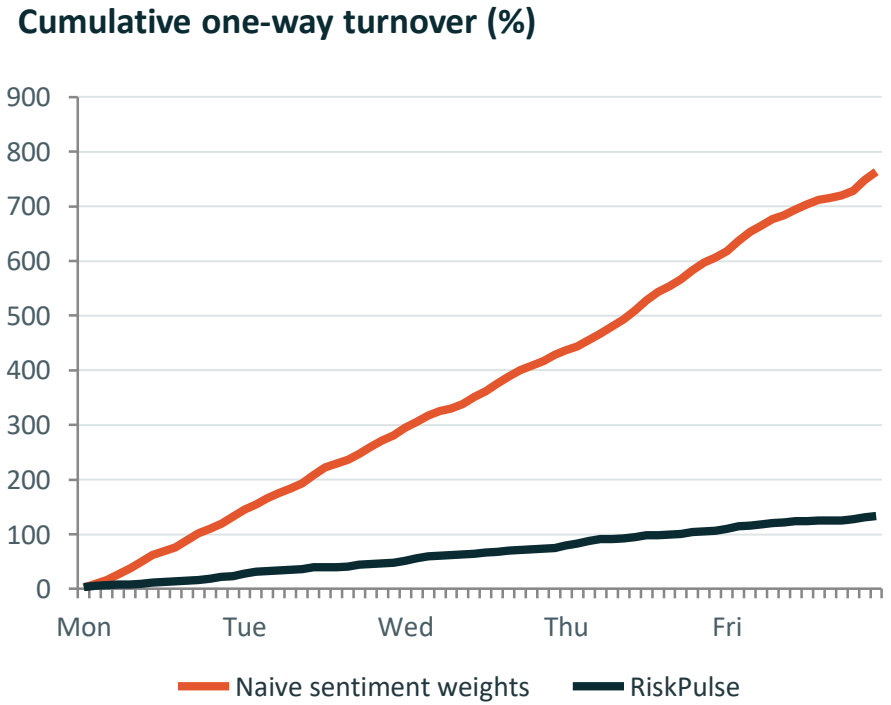
event-class accuracy on a held-out set (n = 48). Sentiment 0.75 vs 0.65 for plain VADER; impact rank correlation 0.76

83%

less trading than naive sentiment weights; cost drag 6.7 vs 38.1 bp over the week



Dotted lines = stress-test triggers: 1 rate hike · 2 Gulf standoff · 3 bank failure · 4 hurricane · 5 ransomware



Domain impact: from headline to P&L in minutes

Thursday, replayed

6.2

10:05 ET · one tweet

Unconfirmed claim of frozen withdrawals from a 412-follower account. Scored, but held below the trigger.

9.5

10:40 ET · wire confirms

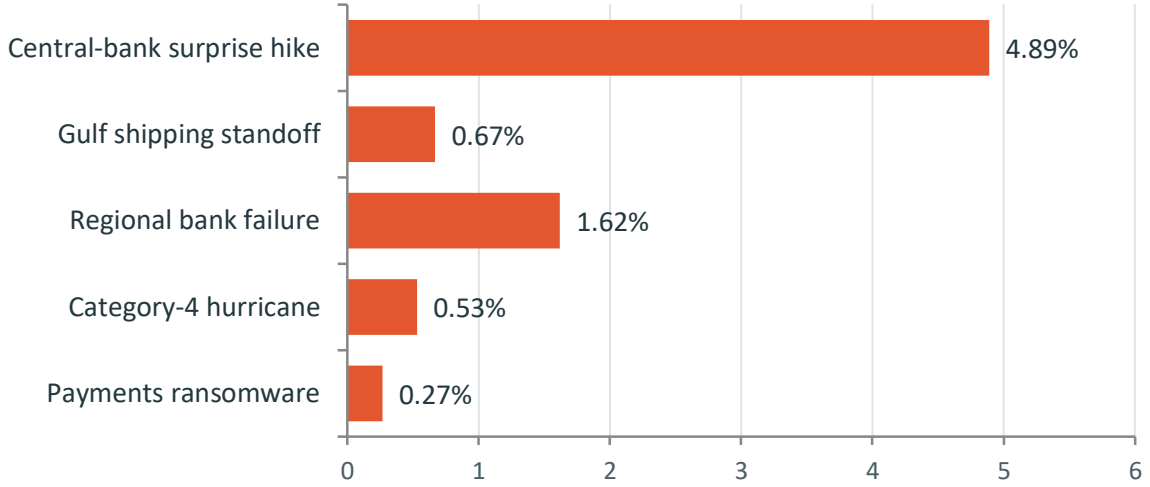
Bankruptcy filing and deposit run reported; two more outlets follow within the hour. A Credit Event stress test fires.

−1.62%

Result on the wholesale book

−$44.6M: rates +32.4, spreads −30.1, credit losses −27.8, equity −19.0. Loan expected loss $21.8M → $49.6M.

Stress loss per triggered event (% of book)



Front office

Sentiment-aware index weights with 83% less turnover.

Risk management

Event-triggered stress tests with factor attribution.

Model risk

Every score ships with evidence, a version and tests.

Limitations and next steps

What I would tell a sceptical risk manager

- Demo data are synthetic and scripted; prices are random, so no alpha is claimed.
- Rule and lexicon NLP misses subtle language. Held-out set is small (n = 48) and self-labelled.
- Impact is a calibrated heuristic: held-out high-impact precision and recall are about 0.5.
- Stress shocks are illustrative sensitivities, linear in impact, not regulatory scenarios.

Next steps

1. Fine-tune FinBERT and an event classifier on Kaggle and GDELT labels; the backend flag already exists.
2. Calibrate impact with event studies on real returns instead of a hand-tuned scale.
3. Move from linear scaling to factor or historical-simulation shocks; add capital and liquidity metrics.
4. Stream with Kafka and add model governance: versioning, drift monitors, audit log.